



Killara
HIGH SCHOOL

STUDENT NUMBER

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MATHEMATICS COURSE

ASSESSMENT TASK NO 2

Date: 7 March 2012

TIME ALLOWED: *Reading time: 5 minutes*
Working time: 3 hours

INSTRUCTIONS:

- This paper consists of 10 Multiple Choice questions and 6 Free Response questions
- ANSWER all 16 questions
- Answer the Multiple Choice questions in the answer sheet provided
- Start each question in a new booklet
- A table of standard integrals is provided at the back of this paper
- All relevant working should be shown.

QUESTION NO	MARK
MC	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15
TOTAL	/100

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

OUTCOMES ASSESSED:

H1 - seeks to apply mathematical techniques to problems in a wide range of practical contexts
H2 - constructs arguments to prove and justify results
H4 - expresses practical problems in mathematical terms based on simple given models
H5 - applies appropriate techniques from the study of calculus and trigonometry to solve problems
H6 - uses the derivative to determine the features of the graph of a function
H7 - uses the features of a graph to deduce information about the derivative
H8 - uses techniques of integration to calculate areas and volumes
H9 - communicates using mathematical language, notation, diagrams and graphs

This assessment task constitutes 25% of the Higher School Certificate Course Assessment.

Section 1

10 marks

Attempt questions 1 – 10

Allow 15 minutes for this section

Use multiple choice answer sheet for Questions 1 – 10

1. The quadratic equation with 5 and -2 as roots is:

(A) $x^2 + 5x - 2 = 0$

(B) $x^2 - 2x + 5 = 0$

(C) $x^2 + 3x - 10 = 0$

(D) $x^2 - 3x - 10 = 0$

2. The domain of $y = \frac{1}{\sqrt{x^2 - 4}}$ is:

(A) $-2 < x < 2$

(B) $-2 \leq x \leq 2$

(C) $x < -2$ or $x > 2$

(D) $x \leq -2$ or $x \geq 2$

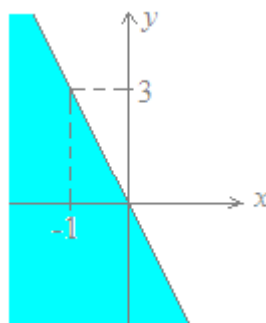
3. The shaded region is best described by which inequality:

(A) $y < -\frac{1}{3}x$

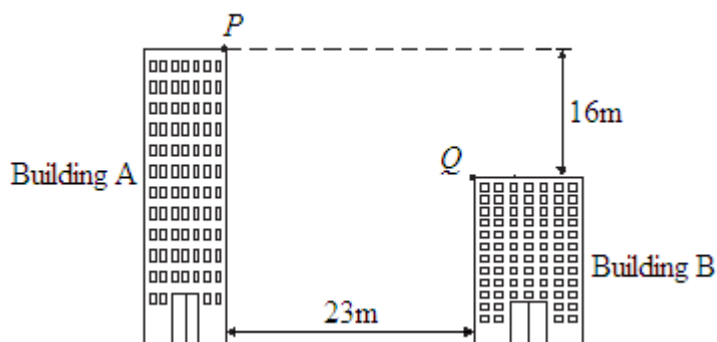
(B) $y \geq -3x$

(C) $y \leq -3x$

(D) $y \leq -\frac{1}{3}x$



4. The point Q on building B is visible from the point P on building A , as shown in the diagram above. Building A is 16m taller than building B .



The horizontal distance between P and Q is 23 metres.

The angle of depression of Q from P is closest to

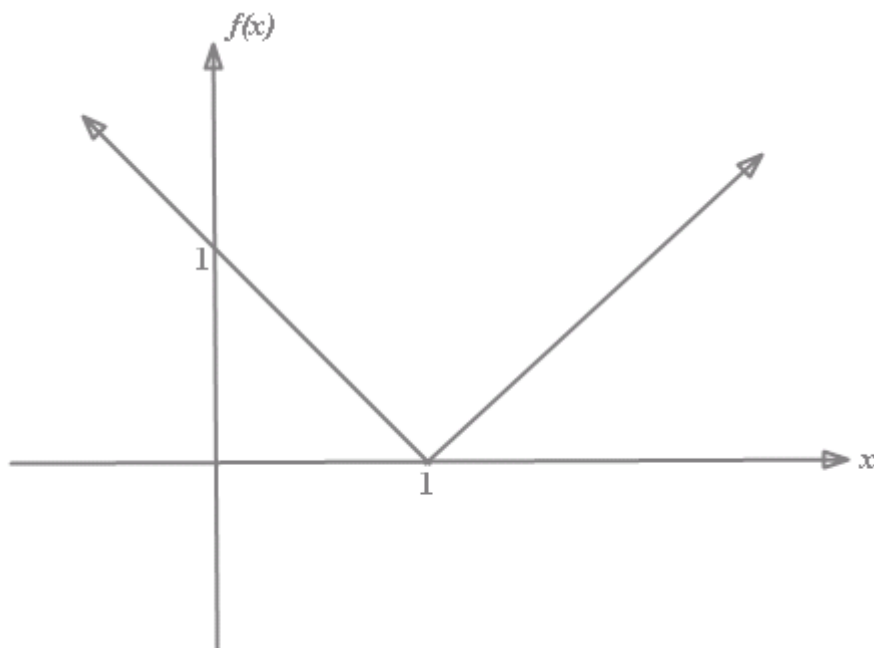
- (A) 35°
 - (B) 41°
 - (C) 44°
 - (D) 46°
5. If a curve is always increasing, which of the following must occur for all values of x ?

- (A) $\frac{dy}{dx} < 0$
- (B) $\frac{dy}{dx} > 0$
- (C) $\frac{d^2y}{dx^2} < 0$
- (D) $\frac{d^2y}{dx^2} > 0$

6. Which of the following technique would you feel to be the best approach to

evaluate $\int_0^2 \sqrt{4-x^2} \, dx$?

- (A) Use substitution and integrate with respect to x
 - (B) Use Trapezoidal rule
 - (C) Use Simpson's rule
 - (D) Use geometrical formula
7. The graph of the function $y = f(x)$ is given below.



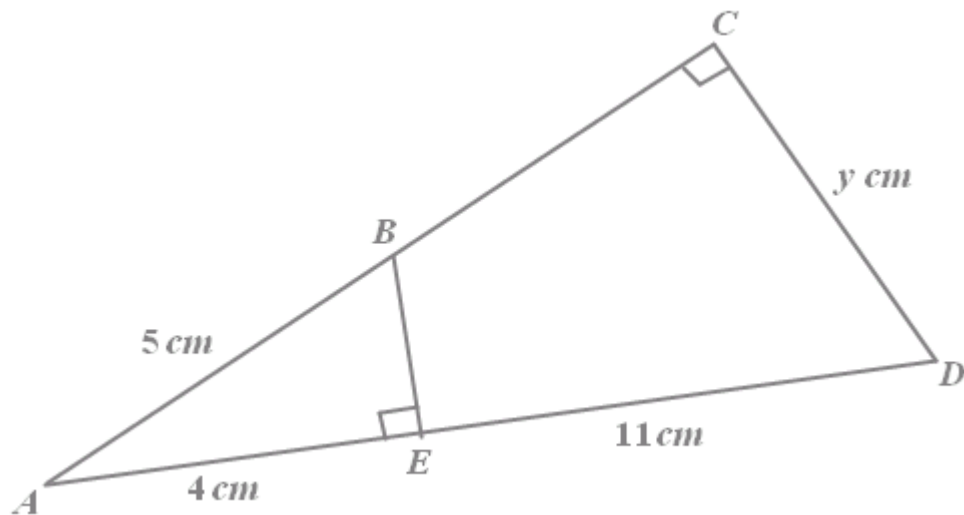
This function is not differentiable at $x = 1$ because

- (A) $y = f(x)$ is not defined at $x = 1$
- (B) $\lim_{x \rightarrow 1} f'(x) \rightarrow 0$
- (C) $\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^+} f'(x) \rightarrow \infty$
- (D) $\lim_{x \rightarrow 1^-} f'(x) \neq \lim_{x \rightarrow 1^+} f'(x)$

8. $\int_{-a}^a f(x) dx = 0$, then

- (A) $f(x)$ is a constant
- (B) $f(x)$ is odd
- (C) $f(x)$ is even
- (D) $f(x)$ cannot be integrated

9.

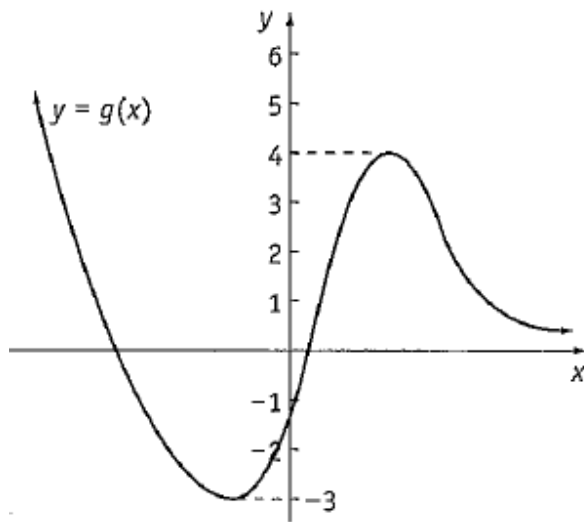


In the diagram given above, the value of y is:

- (A) 8
- (B) 9
- (C) 6
- (D) 4

10. The diagram shows the graph of $y = g(x)$.

Which of the following show all values of k for which $g(x) = k$ have three solutions?



- (A) $-3 \leq k \leq 4$
- (B) $-3 \leq k \leq 0$
- (C) $1 < k < 4$
- (D) $0 < k < 4$

End of Multiple Choice Questions

Section 2

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

All necessary working should be shown for each question

Question 11 (15 marks) Use a SEPARATE working booklet

Marks

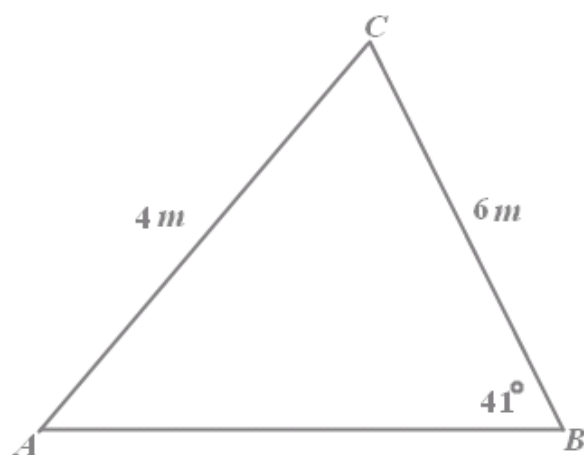
- a) Evaluate $\sum_{k=1}^4 k^2$ **1**
- b) Factorise fully: $3x^2 + x - 2$ **2**
- c) Solve the equation $\frac{x}{3} - \frac{x-1}{2} = \frac{1}{2}$ **3**
- d) Find the exact value of $\tan 120$ **2**
- e) Find the values of x for which $|2x - 1| \leq 5$ **2**

Question 11 continued on page 8

f) (i) Express $3\sqrt{48} - 2\sqrt{27}$ in its simplest surd form. 2

(ii) Hence, find p given that $3\sqrt{48} - 2\sqrt{27} = \sqrt{p}$ 1

g) Find $\angle BAC$ in the following triangle to the nearest degree. 2

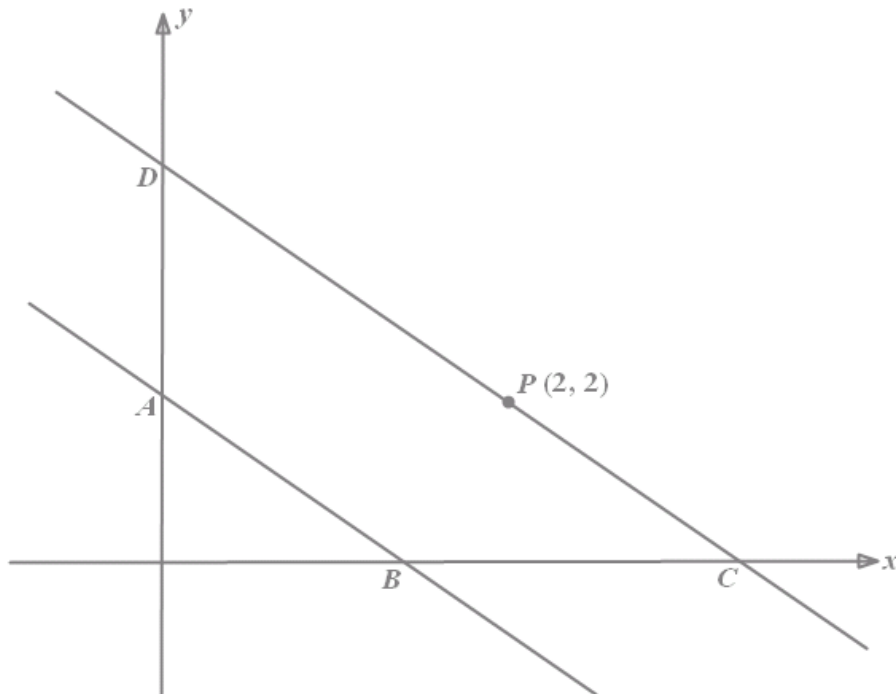


NOT TO SCALE

End of Question 11

- a) The diagram shows the straight line AB with equation $x + 2y = 2$.

Copy or trace the diagram on to your answer booklet.



- (i) DC is parallel to AB and passes through $(2, 2)$.

Show that the equation of DC is $x + 2y = 6$

2

- (ii) Calculate the perpendicular distance from P to AB .

1

- (iii) Show that the distance AB is $\sqrt{5}$.

3

- (iii) Calculate the area of the trapezium $ABCD$

3

- b) M is the midpoint of $A(5, k)$ and $B(11, 8)$.

3

Find k , if M lies on the line $x - 2y = 4$.

- c) Consider the parabola with equation $y^2 = 4x - 8$. Find:

- (i) the coordinates of the vertex

1

- (ii) the coordinates of the focus.

2

End of Question 12

Question 13 (15 marks) Use a SEPARATE working booklet

Marks

a) Differentiate:

(i) $y = \frac{3x+2}{4x-3}$ 2

(ii) $y = x\sqrt{x}$ 2

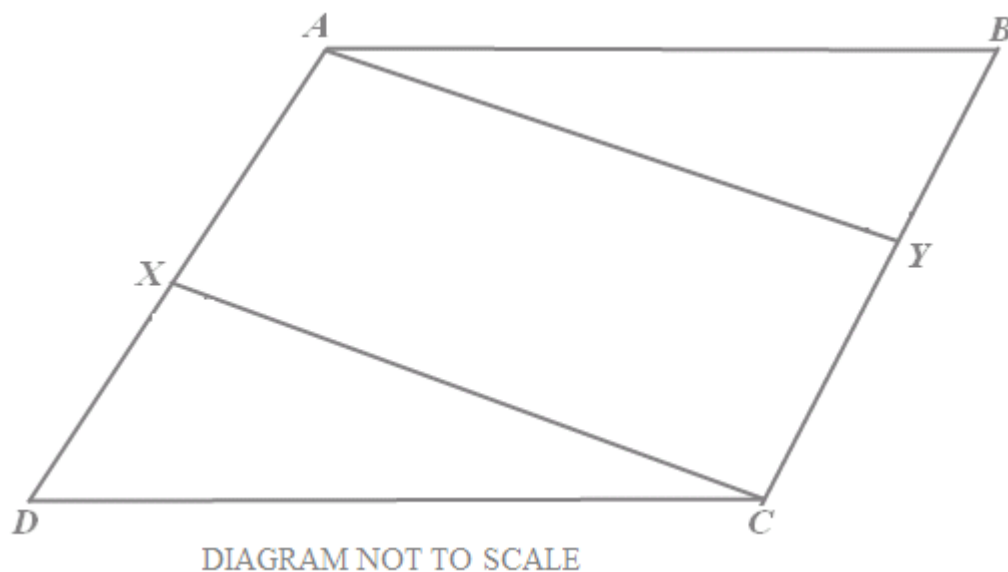
(iii) $y = (2x+1)(5x-3)^4$ 2

b) Derive the equation of the tangent to the curve $y = 2x^2 + 5$ at the point where $x = 2$. 2

c) Sketch the continuous curve $y = f(x)$ which satisfies the following conditions. 3

- o $f'(x) < 0$ for $x < -1$
- o $f'(-1) = f'(2) = 0$
- o $f''(0) = 0$
- o $f''(x) < 0$ for $0 < x \leq 5$
- o $f(4) = 1$

d)



$ABCD$ is a rhombus. X and Y lie on AD and BC such that $XC \perp AD$ and $AY \perp BC$.

Copy or trace the diagram on to your answer booklet.

(i) Prove that $\triangle ABY \equiv \triangle DXC$. 2

(ii) If the side length of the rhombus is 10 cm and $\angle DAB = 120^\circ$, find the length of diagonal AC . 2

End of Question 13

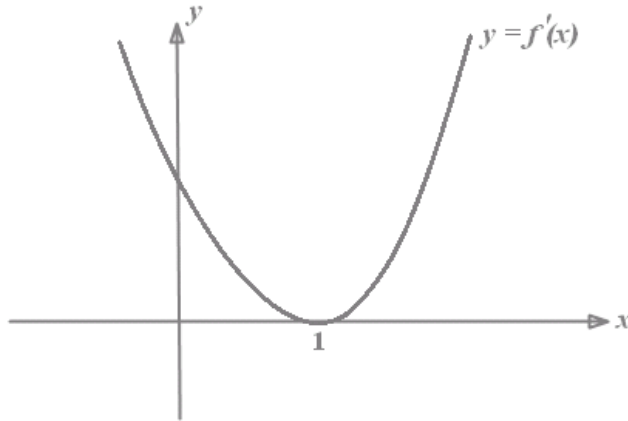
Question 14 (15 marks) Use a SEPARATE working booklet

Marks

- a) Consider the function $f(x) = 9x - 3x^2 - x^3$.
- (i) Find the coordinates of the two stationary points. **2**
- (ii) Determine the nature of the stationary points. **2**
- (iii) Show that there is a point of inflection at $(-1, -11)$. **2**
- (iv) Sketch the curve $y = f(x)$ showing the stationary points and the point of inflection. **1**
- (v) For what values of x is the curve concave up? **1**

Question 14 continues on page 14

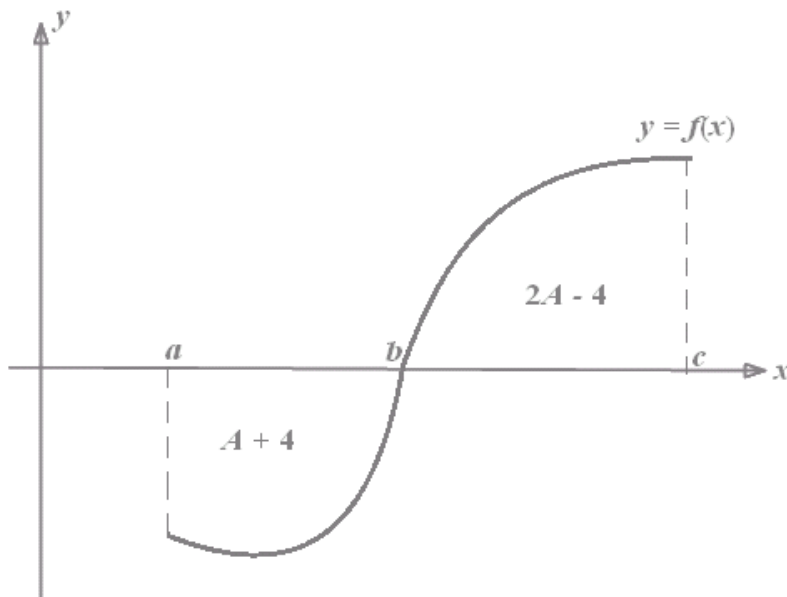
- b) Consider the function $y = f(x)$, whose gradient function is shown as the parabola in the sketch below.



- (i) Comment on the sign of $\frac{dy}{dx}$ for all $x \neq 1$ 1
- (ii) There is a stationary point (a horizontal point of inflection) at $x = 1$.
Justify this statement with reference to the graph. 2
- (iii) Copy the graph above onto your answer sheet.
On the same set of axes, sketch the graph of $y = f''(x)$. 1
- c) The function $y = ax^2 + bx + c$ has the gradient function $\frac{dy}{dx} = 4x + 2$. 3
The function has a minimum value of 1. Find the values of a , b and c .

End of Question 14

- a) In the diagram below, the area between the curve $y = f(x)$, the x -axis and the line $x = a$ is equal to $(A + 4)$ unit². The area between the curve $y = f(x)$, the x -axis and the line $x = c$ is equal to $(2A - 4)$ unit².



Use the information to evaluate $\int_a^c f(x) dx$.

1

- b) Find:

(i) $\int \frac{4x^3 - 3x^2 + 5}{x^2} dx$

2

(ii) $\int_0^1 (2 - 5x)^4 dx$

2

Question 15 continues on page 16

c) The function $f(x)$ is defined by the rule $f(x) = 5x\sqrt{x+1}$ in the domain $-1 \leq x \leq 1$

(i) Show that the derivative of $f(x)$ is $\frac{15x+10}{2\sqrt{x+1}}$. 1

(ii) Use the derivative of $f(x)$ to find the turning point of $f(x)$ and determine its nature. Answer to one decimal place. 2

(iii) Hence, draw a rough sketch of $f(x)$, showing clearly the turning points, and the values at the endpoints of the domain. 2
(You do not have to scale the diagram accurately and do not find the points of inflection)

(iv) the area between the x-axis and the section of the curve $y = f(x)$ from $x = -1$ to $x = 0$ is rotated about the x-axis. 3
 Find the volume of the solid of revolution.

(v) Copy and complete the table below correct to *one decimal place* and apply Simpson's rule with five function values to give an

estimate of $\int_{-1}^1 5x\sqrt{x+1} dx$.

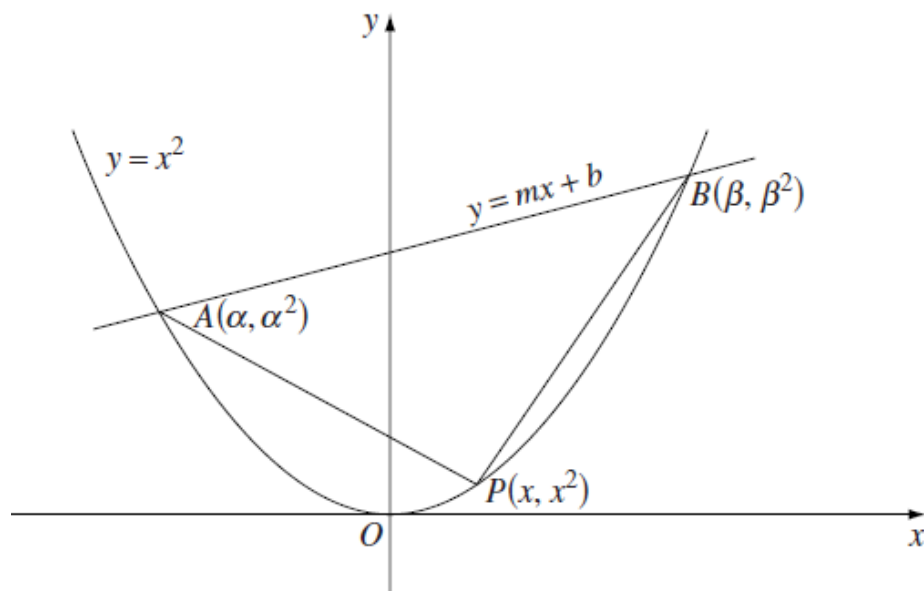
x	-1	-0.5	0	0.5	1
$f(x)$					

End of Question 15

Question 16 (15 marks) Use a SEPARATE working booklet

Marks

- a) The parabola $y = x^2$ and the line $y = mx + b$ intersect at the points $A(\alpha, \alpha^2)$ and $B(\beta, \beta^2)$ as shown in the diagram.



- (i) Explain why $\alpha + \beta = m$ and $\alpha\beta = -b$. 1

- (ii) Show that $(\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2 = (\alpha - \beta)^2 [1 + (\alpha + \beta)^2]$ 2

- (iii) Hence, Show that the distance 2

$$AB = \sqrt{(m^2 + 4b)(1 + m^2)}$$

- (iv) The point $P(x, x^2)$ lies on the parabola between A and B.

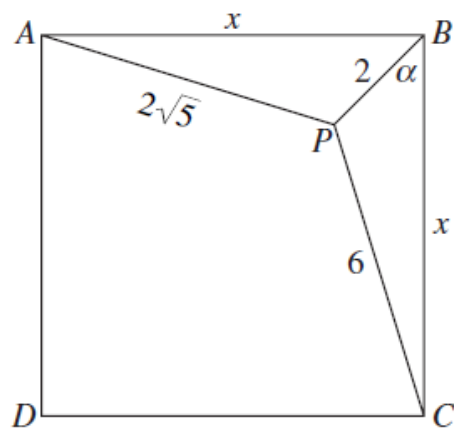
Show that the area of triangle ABP is given by

2

$$\frac{1}{2}(mx - x^2 + b)\sqrt{m^2 + 4b}$$

Question 16 continues on page 18

- b) $ABCD$ is a square with side x cm, with P a point within the square such that $PC = 6$ cm, $PB = 2$ cm and $AP = 2\sqrt{5}$ cm.
Let $\angle PBC = \alpha$.



NOT TO SCALE

- (i) By using triangle PBC show that $\cos \alpha = \frac{x^2 - 32}{4x}$ 1
- (ii) By considering triangle PBA , show that $\sin \alpha = \frac{x^2 - 16}{4x}$ 2
- (iii) Hence, or otherwise, show that the value of x is a solution of 2

$$x^4 - 56x^2 + 640 = 0$$
- (iv) Find x . Give reason for your answer. 3

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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Multiple Choice Answer Sheet

Completely fill the response oval representing the most correct answer.

1. A ○ B ○ C ○ D ○
2. A ○ B ○ C ○ D ○
3. A ○ B ○ C ○ D ○
4. A ○ B ○ C ○ D ○
5. A ○ B ○ C ○ D ○
6. A ○ B ○ C ○ D ○
7. A ○ B ○ C ○ D ○
8. A ○ B ○ C ○ D ○
9. A ○ B ○ C ○ D ○
10. A ○ B ○ C ○ D ○

2012 HSC Half yearly marking scheme

1. D	2. C	3. C	4. A	5. B
6. D	7. D	8. B	9. B	10. D

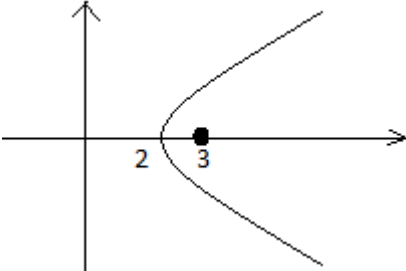
Question 11

a) $\sum_{k=1}^4 k^2 =$ $1 + 4 + 9 + 16 = 30$	1 mark: correct answer	
b) $3x^2 + x - 2$ $= (x+1)(3x-2)$	1 mark	
c) $\frac{x}{3} - \frac{x-1}{2} = \frac{1}{2}$ $\frac{2x}{3 \times 2} - \frac{3(x-1)}{2 \times 3} = \frac{1 \times 3}{2 \times 3}$ $2x - 3(x-1) 2x = 3$ 1 mark $2x - 3x + 3 = 3$ 1 mark $-x = 0$ $x = 0$	1 mark: manipulates equation to remove the denominators 1 mark: removes brackets and simplifies correctly 1 mark: correctly solves the equation	
d) $\tan 120 = -\tan 60$ $= -\sqrt{3}$	1 mark: expresses in terms of the acute angle $-\tan 60$ 1 mark: gives correct exact value	
e) $ 2x-1 \leq 5$ $-5 \leq 2x-1 \leq 5$ 1 mark $-4 \leq 2x \leq 6$ $-2 \leq x \leq 3$ 1 mark	1 mark: removes the absolute value sign and gives the correct inequalities. 1 mark: correct solution	
f) (i) $3\sqrt{48} - 2\sqrt{27}$ $= 3 \times 4\sqrt{3} - 2 \times 3\sqrt{3}$ 1 mark $= 12\sqrt{3} - 6\sqrt{3}$ $= 6\sqrt{3}$ 1mark	1 mark: correctly simplifies the surds $\sqrt{48}$ and $\sqrt{27}$ 1 mark: correct answer	
f)(ii) $3\sqrt{48} - 2\sqrt{27} = \sqrt{p}$ using (i) $6\sqrt{3} = \sqrt{p}$ $\sqrt{108} = \sqrt{p}$ $\therefore p = 108$	1 mark: correct answer	

$1(g) \quad \frac{\sin \theta}{6} = \frac{\sin 41}{4}$ $\sin \theta = \frac{\sin 41}{4} \times 6 = 0.98408\dots$ $\Theta = \sin^{-1}(0.98408\dots)$ $= 79.765\dots$ $= 80^\circ$	1 mark: correctly applies the Sine rule 1 mark: correct answer rounded to the nearest degrees from the calculation of the sine rule.	
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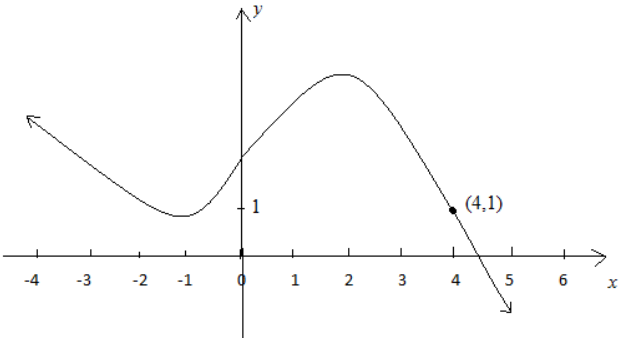
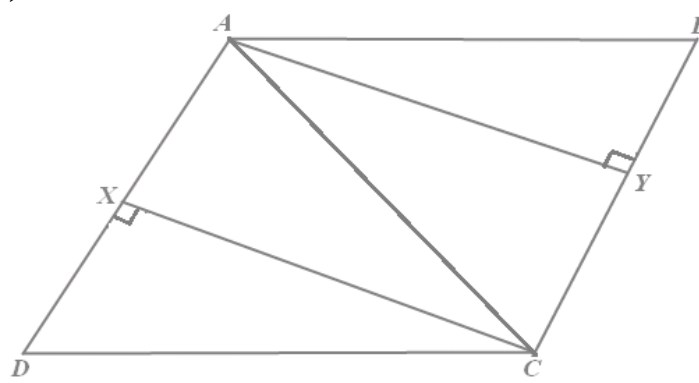
Question 12

a)(i) DC $\parallel$ AB $\therefore$ DC has equation $x + 2y + k = 0$ 1 mark (2, 2) lies on it. $2 + 2 \times 2 + k = 0$ $\therefore k = -6$ $\therefore$ DC has equation $x + 2y - 6 = 0$ $x + 2y = 6$ 1 mark	1 mark: for giving the correct gradient of DC 1 mark: substitutes (2, 2) to get the correct equation	
b) P(2, 2) $x + 2y - 2 = 0$ $\perp$ distance = $\left \frac{2 + 2 \times 2 - 2}{\sqrt{1^2 + 2^2}} \right$ $= \frac{4}{\sqrt{5}}$	1 mark: correct solution	
c) $x + 2y - 2 = 0$ For A, $x = 0 \therefore 2y = 2$ $y = 1$ (0, 1) 1 mark For B, $y = 0 \therefore x = 2$ (2, 0) 1 mark $AB = \sqrt{1^2 + 2^2} = \sqrt{5}$ 1 mark	1 mark: correct y intercept 1 mark: correct x intercept 1 mark: uses the distance formula or Pythagoras' theorem to evaluate AB	
d) Finding length of CD $x + 2y - 6 = 0$ For D, $x = 0 \therefore y = 3$ For C, $y = 0 \therefore x = 6$ $CD = \sqrt{3^2 + 6^2} = \sqrt{45}$ $= 3\sqrt{5}$	1 mark: finds length of CD (giving $\sqrt{45}$ is enough)	

Area of trapezium = $\frac{1}{2} \times \frac{4}{\sqrt{5}} \times (\sqrt{5} + 3\sqrt{5})$ $= 8 \text{ unit}^2 \quad \mathbf{1 \text{ mark}}$	1 mark: substitutes into correct formula 1 mark: evaluates area correctly.	
e) $M\left(\frac{11+5}{2}, \frac{k+8}{2}\right) = \mathbf{1 \text{ mark}}$ $M\left(8, \frac{k+8}{2}\right)$ This point lies on $x - 2y = 4$ $8 - 2 \times \left(\frac{k+8}{2}\right) = 4 \quad \mathbf{1 \text{ mark}}$ $8 - k - 8 = 4$ $\therefore -k = 4$ $\therefore k = -4 \quad \mathbf{1 \text{ mark}}$	1 mark: uses the midpoint formula to get the correct expression for midpoint 1 mark: substitutes the coordinates of M in to the equation 1 mark: solves for k.	
c)(i) $y^2 = 4x - 8$ $y^2 = 4(x - 2)$ Vertex (2,0) $\mathbf{1 \text{ mark}}$ (ii) $4a = 4$ $a = 1 \quad \mathbf{1 \text{ mark}}$ focus (3, 0) $\mathbf{1 \text{ mark}}$ 	1 mark: expresses the equation in the form $(y - k) = 4a(x - h)$ and gives the correct vertex (ii) 1 mark: calculates the focal length correctly 1 mark: calculates the correct focus	

Question13

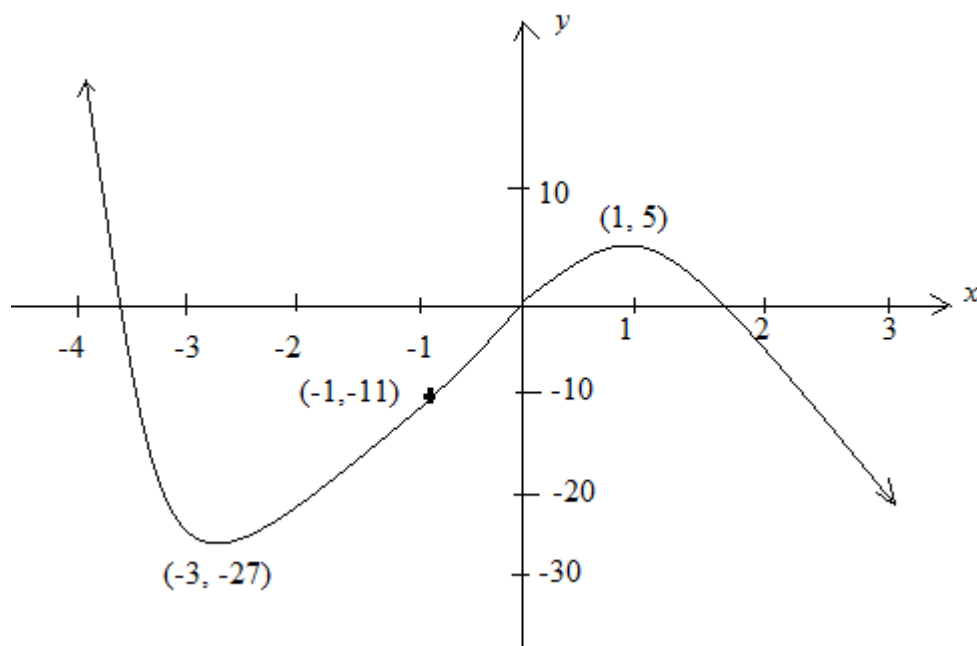
a)(i) $y = \frac{3x+2}{4x-3}$ $u = 3x+2 \quad u' = 3$ $v = 4x-3 \quad v' = 4$ 1 mark $\frac{dy}{dx} = \frac{(4x-3) \times 3 - (3x+2) \times 4}{(4x-3)^2}$ $= \frac{12x-9-12x-8}{(4x-3)^2} = \frac{-17}{(4x-3)^2}$	1 mark: correctly differentiates $3x+2$ and $4x-3$. 1 mark: applies the quotient rule correctly	
(ii) $y = x\sqrt{x} = x^{\frac{3}{2}}$ 1 mark $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}}$ 1 mark	1 mark: writes $x\sqrt{x}$ in index form 1 mark: correctly differentiates	
(iii) $y = (2x+1)(5x-3)^4$ $u = 2x+1 \quad u' = 2$ $v = (5x-3)^4$ $v' = 4(5x-3)^3 \times 5$ 1 mark $\frac{dy}{dx} = (2x+1) \times 4(5x-3)^3 \times 5 +$ $(5x-3)^4 \times 2$ $= 2(5x-3)^3 [20x+10+5x-3]$ $= 2(5x-3)^3 [25x+7]$	1 mark: applies chain rule to differentiate $(5x-3)^4$ 1 mark: applies product rule	
b) $y = 2x^2 + 5$ $\frac{dy}{dx} = 4x$ When $x = 2$, $y = 2 \times 2^2 + 5 = 13$ $(2, 13)$ $\frac{dy}{dx}$ at $x = 2$, is $4 \times 2 = 8$ 1 mark Equation of the tangent is $y - 13 = 8(x - 2)$ $8x - y - 3 = 0$ is the equation of the tangent 1 mark	1 mark: correctly evaluates the gradient of the tangent ($\frac{dy}{dx} = 8$) 1 mark: uses their gradient and the point $(2, 13)$ to find the equation of the line.	

c) 	1 mark: correct shape for $x < 0$. Correct gradient for $x < -1$, stat. point at $x = 1$, correct concavity required 1 mark: concave down for $0 < x \leq 5$, including point of inflection at $x = 0$ 1 mark: max. turning point at $x = 2$ and the point $(4, 1)$ is plotted.	
d)  (i) $AB = CD$ (sides of a rhombus are equal) $\angle AYB = 90^\circ$ $\angle AXC = 90^\circ$ ($XC \perp AD$ and $AY \perp BC$) 1 mark $\angle B = \angle D$ (opposite angles of a rhombus are equal) $\therefore \triangle ABY \equiv \triangle DXC$ (AAS angle-angle side) 1 mark		
(ii) Construct AC. $\angle DAC = 60^\circ$ $\angle ADC = 60^\circ$ (the diagonal bisects the vertices of a rhombus) 1 mark $\triangle ADC$ is equilateral (each of the vertices of the triangle = 60°. <u>Sides opposite equal angles are equal must have this</u>) $\therefore AC = 10\text{cm}$ 1 mark		

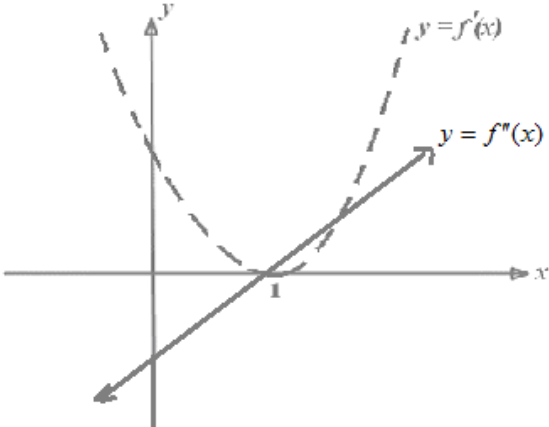
Question 14

a) $f(x) = 9x - 3x^2 - x^3$. $f'(x) = 9 - 6x - 3x^2$. For stationary points, $f'(x) = 0$ $9 - 6x - 3x^2 = 0$ $-3(x^2 + 2x - 3) = 0$ $(x + 3)(x - 1) = 0$ $x = -3, x = 1$ When $x = -3, y = -27$ When $x = 1, y = 5$ Stationary points are $(-3, -27)$ and $(1, 5)$.	1 mark for each of the stationary points									
$f''(x) = -6 - 6x$ When $x = -3, f''(-3) = 12 > 0$ $\therefore$ concave up; Minimum turning point $(-3, -27)$ 1 mark When $x = 1, f''(1) = -6 - 6 < 0$ $\therefore$ concave down;; Maximum turning point $(1, 5)$ 1 mark	1 mark: proves that there is a minimum turning point at $(-3, -27)$ 1 mark: proves that there is a maximum turning point at $(1, 5)$									
(iii) For points of inflection $f''(x) = 0$ $\therefore -6 - 6x = 0$ $x = -1$ When $x = -1, f(x) = -11$ Checking for concavity: <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td></tr><tr><td>$f''(x)$ $= -6 - 6x$</td><td>6 </td><td>0</td><td>-6 </td></tr></table> Change in concavity and therefore point of inflection at $(-1, -11)$ must have this	x	-2	-1	0	$f''(x)$ $= -6 - 6x$	6 	0	-6 	1 mark: sets $f''(x) = 0$ and finds the possible point of inflection. 1 mark: shows working to establish that there is a change in concavity and verbal explanation is required	
x	-2	-1	0							
$f''(x)$ $= -6 - 6x$	6 	0	-6 							

(iv)

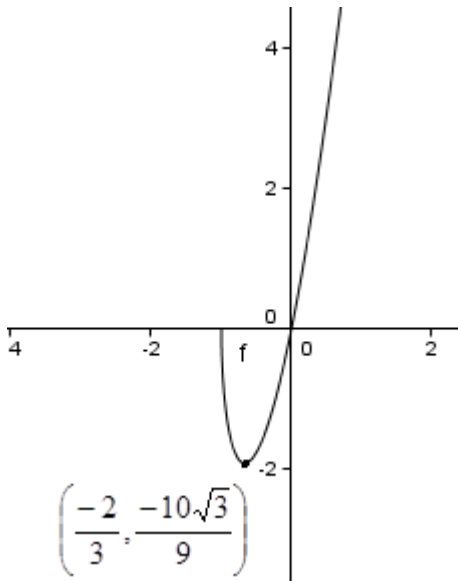


(iv)	1 mark: correct shape with the turning points, point of inflection and the x-intercept shown	
(v) The graph is concave up for $x < -1$	1 mark: correctly writes the region from their graph Or sets $f''(x) < 0$ and solves the inequality	
(b)(i) $\frac{dy}{dx} > 0$ for all $x, x \neq 1$	Correct answer, must say $x \neq 1$ 1 mark	
(ii) $\frac{dy}{dx} = 0$ at $x = 1$ $\therefore$ stationary point. 1 mark $\frac{dy}{dx}$ is positive on both sides of $x = 1$. it does not change sign.  or $\frac{d^2y}{dx^2} = 0$ at $x = 1$; $\frac{d^2y}{dx^2} < 0$ for $x < 1$ and $\frac{d^2y}{dx^2} > 0$ for $x > 1$. Change in concavity, ie point of inflection at $x = 1$.	1 mark for identifying the stationary point at $x = 1$ 1 mark for establishing point of inflection.	

$\therefore x = 1$ is a horizontal point of inflection.		
(iii) 	1 mark for correct diagram of $y = f''(x)$	
c) $y = ax^2 + bx + c$ $\frac{dy}{dx} = 2ax + b = 4x + 2$ Comparing coefficients, $\therefore 2a = 4 \quad a = 2$ 1 mark $b = 2$ 1 mark $\therefore y = 2x^2 + 2x + c$ minimum value of 1. $X = \frac{-b}{2a} = \frac{-1}{2}$ $X = \frac{-1}{2}, y = 1$ $1 = 2\left(-\frac{1}{2}\right)^2 + 2\left(-\frac{1}{2}\right) + c$ $c = \frac{3}{2}$ $\therefore a = 2, b = 2, c = \frac{3}{2}$	1 mark each for a and b for setting $\frac{dy}{dx} = 4x + 2$ and comparing and evaluating coefficients 1 mark: for the value of c. (must show working)	

Question 15

(a) $-(A+4) + (2A-4)$ $= -A - 4 + 2A - 4$ $= A - 8$	1 mark: Subtracts the area from below the x-axis from the area above the x-axis.	
b)(i) $\int \frac{4x^3 - 3x^2 + 5}{x^2} dx$ $= \int 4x - 3 + 5x^{-2} dx$ $= \frac{4x^2}{2} - 3x + \frac{5x^{-1}}{-1} + C$ $= 2x^2 - 3x - \frac{5}{x} + C$	1 mark: performs term by term division correctly 1 mark: correct integration	
b)(ii) $\int_0^1 (2-5x)^4 dx$ $\left[\frac{(2-5x)^5}{5 \times -5} \right]_0^1$ 1 mark $= \frac{-1}{25} \{ (2-5)^5 - (2-0)^5 \}$ $= \frac{-275}{-25} = 11$	1 mark: correct integration 1 mark: correct evaluation	
c) (i) $f(x) = 5x\sqrt{x+1}$ $u = 5x \quad u' = 5$ $v = \sqrt{x+1} \quad v' = \frac{1}{2\sqrt{x+1}}$ $f'(x) = \frac{5x}{2\sqrt{x+1}} + 5\sqrt{x+1}$ $= \frac{5x + 2(x+1)}{2\sqrt{x+1}}$ $= \frac{15x+10}{2\sqrt{x+1}}$	1 mark: correct proof. Every step of working is required.	
c)(ii) $f'(x) = 0$ $\frac{15x+10}{2\sqrt{x+1}} = 0$ $15x + 10 = 0$ $\therefore x = \frac{-2}{3}$ 1 mark when $x = \frac{-2}{3}$, $y = \frac{-10\sqrt{3}}{9} = 1.9$	1 mark: sets $f'(x) = 0$ and solves to get the x-value of the turning point	

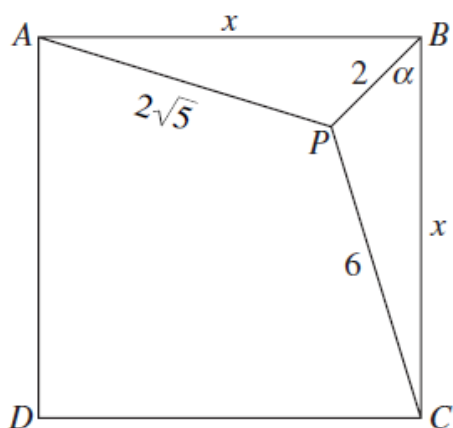
<table><tr><td>x</td><td>-0.8</td><td>$-\frac{2}{3}$</td><td>0</td></tr><tr><td>$f'(x)$</td><td>$-\sqrt{5}$</td><td>0</td><td>5</td></tr></table> $\therefore$ minimum turning point at $(-\frac{2}{3}, 1.9)$	x	-0.8	$-\frac{2}{3}$	0	$f'(x)$	$-\sqrt{5}$	0	5	1 mark: Uses gradient test or second derivative to determine the nature of the stationary point. <i>Must give the turning point as a coordinate pair</i>	
x	-0.8	$-\frac{2}{3}$	0							
$f'(x)$	$-\sqrt{5}$	0	5							
(iii) 										
(iv) Volume = $\int_{-1}^0 \pi \times 25x^2(x+1)dx$ 1 mark $= 25\pi \int_{-1}^0 x^3 + x^2 dx$ $= 25\pi \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_{-1}^0$ 1 mark $= \left 25\pi \left(0 - \left(\frac{1}{4} - \frac{1}{3} \right) \right) \right $ $= \frac{25\pi}{12} \text{unit}^3$ 1 mark	1 mark: correct formula to evaluate the area 1 mark: correctly integrates 1 mark: correct evaluation. Unit required.									

(iv)														
<table><tr><td>x</td><td>-1</td><td>-0.5</td><td>0</td><td>0.5</td><td>1</td></tr><tr><td>$f(x)$</td><td>0</td><td>-1.8</td><td>0</td><td>3.06</td><td>7.07</td></tr></table>	x	-1	-0.5	0	0.5	1	$f(x)$	0	-1.8	0	3.06	7.07	1 mark: correctly completes the table.	
x	-1	-0.5	0	0.5	1									
$f(x)$	0	-1.8	0	3.06	7.07									
$\int_{-1}^1 5x\sqrt{x+1} \, dx \approx$ $\frac{0.5}{3}(0 + 4 \times -1.8 + 2 \times 0 + 4 \times 3.06 + 7.07)$ $= -2.018 \dots \approx -2.0 \text{ (1 d.p.)}$	1 mark: correct substitution and evaluation into the correct formula													

Question 16

Question 10		
a)(i) From the diagram, the roots of the simultaneous equations $y = x^2$ and $y = mx + b$ are $x = \alpha$ and $x = \beta$. Now $x^2 = mx + b$ $x^2 - mx - b = 0$ sum of roots $(\alpha + \beta) = \frac{-b}{a}$ $\alpha + \beta = \frac{-(-m)}{1} = m$ 1 mark Product of roots $(\alpha\beta) = \frac{c}{a}$ $= \frac{-b}{1} = -b$ 1 mark	1 mark: develops the correct quadratic equation, $x^2 - mx - b = 0$ and gives sum of roots $(\alpha + \beta) = \frac{-b}{a}$ and product of roots $(\alpha\beta) = \frac{c}{a}$	
(ii) $(\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2 =$ $(\alpha - \beta)^2 + [(\alpha + \beta)(\alpha - \beta)]^2$ $= (\alpha - \beta)^2 + (\alpha + \beta)^2 (\alpha - \beta)^2$ 1 mark $= (\alpha - \beta)^2 [1 + (\alpha + \beta)^2]$ 1 mark	1 mark: correctly factorises and expands $(\alpha^2 - \beta^2)^2$ as $(\alpha + \beta)^2 (\alpha - \beta)^2$ 1 mark: correctly factorises to get the required result	
(iii) By distance formula $AB = \sqrt{(\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2}$ $= \sqrt{(\alpha - \beta)^2 [1 + (\alpha + \beta)^2]}$ Now $(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$ $= \alpha^2 + 2\alpha\beta + \beta^2 - 4\alpha\beta$ $= (\alpha + \beta)^2 - 4\alpha\beta$ $= m^2 - 4(-b)$ $= m^2 + 4b$	2 marks: for correct solution Award only 1 mark if: Writes $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ Or Application of distance formula and uses the result from (ii)	

$AB = \sqrt{(\alpha - \beta)^2 [1 + (\alpha + \beta)^2]}$ $= \sqrt{(m^2 + 4b) [1 + (m)^2]}$ $= \sqrt{(m^2 + 4b)(1 + m^2)}$		
(iv) perpendicular distance from $P(x, x^2)$ to the line $mx - y + b = 0$ is given by: $h = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ mx - x^2 + b }{\sqrt{m^2 + 1}}$ $= \frac{mx - x^2 + b}{\sqrt{m^2 + 1}} \text{ (from the diagram, } mx + b > x^2 \text{)}$ Area of triangle ABP, $= \frac{1}{2} \times AB \times \text{perp. distance from } AB$ $= \frac{1}{2} \times \sqrt{(m^2 + 4b)(1 + m^2)} \times \frac{mx - x^2 + b}{\sqrt{m^2 + 1}}$ $= \frac{1}{2} \times \sqrt{(m^2 + 4b)} \times \sqrt{(1 + m^2)} \times \frac{mx - x^2 + b}{\sqrt{m^2 + 1}}$ $= \frac{1}{2} (mx - x^2 + b) \sqrt{m^2 + 4b}$	2 marks: correct solution Award only 1 mark: attempts to calculate perpendicular from P to AB or attempts to find the area of the Δ by writing $A = \frac{1}{2}bh$ and applies their base length AB from (iii).	



16(b)(i) In $\triangle PBC$, $\cos \alpha = \frac{PB^2 + CB^2 - PC^2}{2 \times PB \times CB}$ $= \frac{2^2 + x^2 - 6^2}{2 \times 2 \times x}$ $= \frac{x^2 - 32}{4x}$	1 mark: applies cosine rule correctly (Every step of working must be shown)	
(ii) Since $\angle ABC$ is a right angle in a square and $\angle PBC = \alpha$, then $\angle ABP = 90^\circ - \alpha$. Using cosine rule in $\triangle ABP$, $\cos \angle ABP = \frac{x^2 + 2^2 - (2\sqrt{5})^2}{2 \times x \times 2}$ $= \frac{x^2 - 16}{4x}$ But $\cos(90^\circ - \alpha) = \sin \alpha$ $\therefore \sin \alpha = \frac{x^2 - 16}{4x}$	1 mark: Recognises $\angle ABP = 90^\circ - \alpha$ and applies cosine rule correctly 1 mark: Uses the property $\cos(90^\circ - \alpha) = \sin \alpha$ and proves the required result	
(iii) From (i), $\cos \alpha = \frac{x^2 - 32}{4x}$ From (ii), $\sin \alpha = \frac{x^2 - 16}{4x}$ But, $\sin^2 \alpha + \cos^2 \alpha = 1$ $\therefore \left(\frac{x^2 - 16}{4x} \right)^2 + \left(\frac{x^2 - 32}{4x} \right)^2 = 1$ $\therefore x^4 - 64x^2 + 1024 + x^4 - 32x^2 + 256 = 16x^2$ $\therefore 2x^4 - 112x^2 + 1280 = 0$ $\therefore x^4 - 56x^2 + 640 = 0$ That is, X is a solution of $x^4 - 56x^2 + 640 = 0$	2 mark: correct solution with all steps of working Award only 1 mark if uses the results from (i) and (ii) and attempts to eliminate $\sin \alpha$ and $\cos \alpha$.	

(iv) $x^4 - 56x^2 + 640 = 0$ ie. $(x^2 - 16)(x^2 - 40) = 0$ $\therefore x = \pm 4$ or $x = \pm \sqrt{40} = \pm 2\sqrt{10}$ 1 mark But x cm is the length of the side of a square, so $x > 0$. $\therefore x = 4$ or $2\sqrt{10}$ 1 mark If $x = 4$, $\cos \alpha = \frac{4^2 - 32}{4 \times 4} < 0$. But α is an acute angle, so $\cos \alpha > 0$. If $x = 2\sqrt{10}$, $\cos \alpha = \frac{(2\sqrt{10})^2 - 32}{4 \times (2\sqrt{10})} > 0$ $\therefore$ solution is $x = 2\sqrt{10}$. 1 mark	1 mark: Correctly solves the equation 1 mark: Recognises x is the length of a side. $\therefore > 0$ 1 mark: Explains why $x = 4$ cannot be a solution and proves $x = 2\sqrt{10}$ is the only solution	
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